Dhairyakumar shah

London, CR7 7HR | shahdhairya1380@gmail.com | +447901170839 | LinkedIn

A dedicated and analytical data science professional with a strong academic foundation in physics and a Master's degree in Data Science (with Distinction). Skilled in applying machine learning, advanced analytics, and data-driven methodologies to solve complex, real-world problems. With hands-on experience in statistical analysis, computer vision, and research-driven data projects, I bring a unique blend of scientific reasoning and technical expertise. Currently holding a Graduate Route visa in the UK, I am actively seeking opportunities, where I can contribute to innovative, data-centric solutions and continue advancing in a dynamic and collaborative environment.

Skills and Abilities

- Programming Languages: C, C++, Python, Internet of Things,
- Libraries: NumPy, Pandas, Seaborn, TensorFlow, Keras, PyTorch, Matplotlib, and Scikit-learn.
- Machine Learning: Regression, Classification, Clustering, RNN, CNN, NLP, Tranfer Learning
- Database Management: Basic knowledge of SQL and SQL Server and the ability to write queries to fetch data from the database.
- Tools/IDE: Google Colab, Git/GitHub, Jupytor notebook, Tableau, Power BI
- Graphic Design: Basic Level in Canva as well picsart and adobe-photoshop
- Problem Solving: I excel in solving complex problems by applying logical and efficient methodologies on competitive platforms.
- Adaptability: Ability to learn and adapt to new technologies and programming languages.
- Technical Proficiency: Model training, hyperparameter tuning, evaluation metrics, EDA

Education

University of Hertfordshire, London — January 2023 - January 2025
MSc in Data Science with Advanced Research — Distinction

Department of Physics, Gujarat University, Gujarat, India — June 2019 – August 2021
MSc in Physics — First Class

Employments

Customer and Trading Manager – Sainsburys, London — August 2024 - Present

- Inventory Management
- Time management
- Team Leadership & Supervision
- Stock Management
- Colleague skills observation

Trading Assistant – Sainsburys, London. — April 2023 – August 2024

- Excellence Customer Service
- Order Picking and Packing: Selecting items from inventory based on customer orders, ensuring accuracy and attention to detail.
- Communication & Interpersonal Skills: Facilitating clear communication between team members and customers.

IT support – Fedora Solutions (USA- INDIA) — April 2022 – August 2022

- Solving the queries with remote desktop
- Providing extra space in cloud or separating drives
- Providing Networking solutions

Projects

Li-FI Technologies- Transferring and receiving data by Solar panel - India

- With the use of Solar panel, the data is getting transferred by the rays of light which can be used in the solar technologies as well radar systems.

GPU-Accelerated Data Processing and Model Training Using TensorFlow

- Developed an efficient data processing pipeline using TensorFlow's tf.data API to accelerate image processing on GPUs. Built and compiled a Convolutional Neural Network (CNN) based on a predefined model summary, and trained it with various callbacks to log the training process over multiple epochs.
- **Tools Used**: TensorFlow, RAPIDS, Python, GPU Acceleration, tf.data API

Stock Market Prediction of Sainsburys, Using Deep Learning with Keras

- Conducted an in-depth analysis of stock data, focusing on seasonality, volatility, and non-stationary trends. Leveraged Keras for stock data preprocessing, training, and forecasting, using data from Yahoo! Finance's API via yfinance. Critically evaluated the effectiveness of deep learning in stock market prediction by reviewing relevant research literature.
- **Tools Used**: Keras, yfinance, Python, Yahoo! Finance API, Deep Learning

Image Classification with Transfer Learning Using Keras

- Explored transfer learning techniques in image classification by leveraging pre-trained models to improve classification performance on a chosen dataset. Investigated the principles of transfer learning, its significance in the context of image classification, and implemented these methods using Keras.
- **Tools Used**: Keras, Python, Transfer Learning, Pre-trained Models

Time Series Modelling & Forecasting Case Study

- Conducted a comprehensive case study on time series data analysis, involving traditional autoregressive methods, Fourier decomposition, and neural network-based modelling for forecasting.

Image Segmentation with COCO-2017 Dataset

- Conducted an image segmentation project utilizing the COCO-2017 dataset. This individual assignment involved comprehensive data analysis, model training, and outcome evaluation.

Plant disease classification using transfer learning on combined dataset (COMPARISON OF VGG16 AND RESNET50)

- Combined two different datasets with python language in Google colab. The project is based on leaves diseased classification in ML models. It advocates the comparison of VGG16 and ResNet50 with the use of EDA and advanced method of transfer learning.

Technical Skills and Expertise

Data Science & Analytics

- Data Analytics & Statistical Analysis: Proficient in analyzing complex datasets using Python (NumPy, Pandas, Scikit-learn) and Excel, applying statistical techniques for data-driven insights.
- Data Handling & Visualization: Experienced in data manipulation, transformation, and visualization using Matplotlib, Seaborn, and advanced Pandas techniques to extract meaningful trends.
- Data Mining & Pattern Discovery: Expertise in data cleaning, feature engineering, dimensionality reduction, and Knowledge Discovery in Databases (KDD) using Python & SQL to extract valuable insights.

Machine Learning & Artificial Intelligence

- Supervised & Unsupervised Learning: Hands-on experience with classification, regression, clustering, and anomaly detection using Scikit-learn, TensorFlow, and PyTorch.
- Neural Networks & Deep Learning: Strong understanding of artificial neural networks (ANNs), CNNs, RNNs, GANs, Vision Transformers, and transfer learning for tasks like image classification, segmentation, and object detection.
- Time Series Analysis: Proficient in autoregressive models (AR, ARIMA), spectral domain methods (Fourier transforms), and Gaussian Processes for forecasting.
- Bayesian Statistics & Modelling: Skilled in Bayesian inference, Naïve Bayes, and probabilistic modelling for anomaly detection and decision-making.

Applied Research & Development

- Advanced Machine Learning Techniques: Experience with Explainable AI (XAI), AutoML, PCA, and network analysis for model interpretability and automation.
- Data Science Workflow: Strong understanding of end-to-end ML pipelines, from data engineering to model deployment, including Kaggle case studies and industry use cases.
- Large Language Models (LLMs): Knowledge of LLMs, biases, scaling, and risks, with expertise in fine-tuning and adapting models for various applications.

Certifications

- Field Technician Network Storage Certificate
- Indian Association of Physics Teachers
- Gujarat science Academy

References

Dr. Sophie Koudmani
Senior Lecturer of Physics Engineer and Computer Science.
Email: skoudmani@ast.cam.ac.uk

Prof D.H. Gadani
Professor at physics and space science department.
Email: dhgadani@gmail.com